

SECTION 07 16 14

ACRYLIC MODIFIED (FLEXIBLE) CEMENTITIOUS WATERPROOFING

Part 1: General

1.1 SECTION INCLUDES

- A. Furnish all labor, materials, tools and equipment as necessary to perform Acrylic Latex Modified Cement Waterproofing on new and existing structures as shown on drawings and as specified in this section.
- B. Related Sections
 - 1. See section 03300 - Cast-in-Place Concrete

1.2 REFERENCES

- A. ASTM C 109 - Standard Test Method for Compressive Strength of Hydraulic Cement Mortars.
- B. ASTM C 348 - Standard Test Method for Flexural Strength of Hydraulic Cement Mortars.
- C. ASTM C 321 - Standard Test Method for Bond Strength of Chemical-Resistant Mortars.
- D. ASTM E 96 - Standard Test Method for Water Vapor Transmission of Materials.
- E. COE CRD-C 48 - Method of Test for Water Permeability of Concrete; U.S. Army Corps of Engineers.

1.3 SUBMITTALS

- A. General

Submit manufacturer's certification that proposed materials, details and systems as indicated and specified fully comply with manufacturer's details and specifications. If any portion of Contract Documents do not conform to manufacturer's standard recommendations, submit notification of portions of design that are at variance with manufacturer's specifications.
- B. Product Data

Submit manufacturer's literature and installation instructions for each product.

1.4 QUALITY ASSURANCE

- A. Manufacturer Qualifications:
 - 1. Company specializing in marketing or manufacturing products specified in this Section with minimum 10 years documented experience.

B. Installer Qualifications:

1. Acceptable to manufacturer with documented experience on at least 5 projects of similar nature in past 5 years and/or training provided by the product manufacturer.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver and store in a dry area between 40oF (5oC) and 90oF (32oC). Handle and protect from freezing and direct sun light in accordance with manufacturer's instructions.
- B. Deliver materials in manufacturer's unopened containers, fully identified with brand, type, grade, class and all other qualifying information. Provide Material Safety Data Sheets for each product.
- C. Take necessary precautions to keep products clean, dry and free of damage.

1.6 SYSTEM REQUIRMENTS

- A. Coordinate waterproofing installation with other trades.
- B. Provide materials and accessories in timely manner so as not to delay Work.

1.7 PROJECT CONDITIONS

- A. Maintain surfaces to be waterproofed and surrounding air temperature at not less than 5 °C. Apply only when temperatures are steady or rising.
- B. Do not apply materials to frozen or frost-filled surfaces.
- C. Exercise caution when temperatures exceed 35 °C.

1.8 WARRANTY

- A. Comply with provisions of Section 01700.
- B. Manufacturer's Warranty: Manufacturer shall provide standard product warranty executed by authorized company official. Term of warranty shall be 5 years from date of substantial completion.

PART 2 - PRODUCTS

1.1 MANUFACTURERS

- A. Approved manufacturer: Saudi Vetonit Company Ltd.-Saveto
71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia
Tel +966 11 2653334, Fax: +966 11 265 0936,
E-mail: info.Riyadh@saveto.com; Web: www.saveto.com.
- B. Requests for substitutions will be considered only if submitted to the

architect/engineer in writing and must include substantiation of product performance, 10 days prior to the original bid date.

1.2 MATERIALS

- A. Waterproofing Material - Acrylic Modified Cement Waterproofing: Cementitious, two-component, acrylic emulsion based, highly flexible, crack bridging waterproof membrane barrier against positive water pressure, with the following characteristics:

Product:	Vetoproof CM745
Density:	1.85 kg/Ltr approx.
Toxicity - BS6920:	Pass
Chloride content:	Nil
Resistance to positive water pressure:	>8 bar (DIN1048)
Resistance to negative water pressure:	3 bar (DIN1048)
Static crack accommodation:	>1mm
Dynamic crack accommodation:	0.3mm
Abrasion resistance ASTM D4060:	Equivalent to 40 N/mm ² concrete
Chloride Diffusion:	No penetration after 24 months.
CO ₂ Diffusion:	>80 m of concrete.
Bond Strength	Higher than concrete cohesive strength
Volatile Organic Content (VOC):	<10 gm / Liter

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine all construction substrates and conditions under which waterproofing materials are to be installed. Do not proceed with the waterproofing application until unsatisfactory conditions are corrected.
- B. Preparation
1. Remove oil, grease, dirt, loose particles, remains of form oils, water repellents, rust or other coatings by high-pressure water blasting (>3000 psi), wet or dry sand blasting, or other mechanical means to produce surface profile ICRI CSP 3 to 5 for application of waterproofing.
 2. Follow manufacturer's instructions to clean and prepare surfaces and seal cracks and joints.
 3. Voids in concrete substrates: 6 mm diameter and larger, pre-treat with patching compound. Less than 6 mm diameter can be filled with a scratch coat of one-component waterproofing material.

- C. Rinse surfaces to be waterproofed (excluding drywall or similar) with clean water to saturated surface dry (SSD) condition, with no standing water on horizontal surfaces.

3.3 INSTALLATION

- A. Mix two-component waterproofing material in proportions recommended by manufacturer.
- B. Cavity fill, honeycombs & form tie holes:
 - a. Fill voids at cleaned and prepared faulty construction joints, cracks, form tie holes with patching compound in mortar consistency flush to surface.
 - b. Laminate patching compound in 2 to 3 layers as per manufacturer's instructions for larger spalled or honeycombed areas.
- C. Taping horizontal and vertical construction joints and cracks (positive side waterproofing only):

Install joint and crack sealing tape, embedded in waterproofing material as follows:

 - 1. Apply two-component waterproofing material by brush in a six to seven inch (15 – 18 cm) wide strip coat centered over all joints, cracks, penetrations and changes of plane to be taped.
 - 2. While this coat is still wet, unroll joint sealing tape into the coating and apply a coat of two component waterproofing material over the tape, smoothing out wrinkles and fish mouths.
- D. Sealing around PVC pipe penetrations:
 - 1. Abrade (sand) PVC pipes and degrease with isopropanol or acetone.
 - 2. Place sealing gasket over pipe and mark size of penetration, then cut out necessary opening (penetration).
 - 3. Apply one prime coat two-component waterproofing material over concrete and exposed PVC pipe.
 - 4. While this coat is still wet, immediately place and firmly press sealing gasket into the coating and cover it with a top coat of two-component waterproofing material, after prime coat has set-up and sufficiently hardened.
- E. Positive Side Waterproofing.

Apply two-component waterproofing material in quantities and number of coats as per manufacturer's specifications and recommendations:

 - 1. Apply at 2mm total thickness for all standard applications (i.e. water Resvoirs, Foot traffic, balconies (non-tiled), etc.
- F. Application Consideration
 - 1. Apply, using stainless steel trowel, Tampico brush, short nap roller, or appropriate compressed-air spray equipment.

2. Apply only when surface and ambient temperatures are 5 °C and rising. At high temperatures 35°C and above, protect application from direct sun and wind to prevent premature surface drying and shrinkage cracks. Apply material in two coats minimum.
3. Application thickness should not exceed 3 mm.
4. If needed, such as in zones posed to movement or cracking, plaza decks, etc., the waterproofing material can be additionally reinforced with a reinforcing mesh (supplied by waterproofing manufacturer), embedded between two waterproofing layers.
5. Do not bridge cracks greater than 1.5 mm.
6. Bridge dynamic cracks or joints with elastomeric joint sealing tape, as supplied by waterproofing manufacturer.

3.4 Curing

- A. Follow manufacturer's general instructions for curing and hardening of waterproofing material. Do not use water for curing. Waterproofing material is self-curing.
- B. Protect surfaces from rain, frost and premature dehydration.

3.5 TESTING OF WATER INCLUDING STRUCTURES

- A. Following application and completion of related work, as required, but well prior to completion of entire project, fill tanks to capacity and allow to stand not less than 3 days. Fill larger structures at a uniform rate not greater than 2 m in 24 hours. The temperature of the fill water shall be plus or minus 10 degrees F of the ambient air and/or the tank structure at the time of filling. Extreme caution is urged if the temperature is greater than 10 degree F. Should leakage occur after this period, drain tanks to perform repairs. Notify Owner prior to draining tanks.

3.6 ACCEPTANCE

- A. Remove left over materials and any foreign material resulting from the work from the site.
- B. Clean adjacent surfaces and materials.